

Supplementary Material for “Fused Exact SwitchGLOBE”

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Abstract

This supplement documents experimental provenance, metric semantics, acceptance gates, negative results, and the complete 20-figure latency and reliability analysis supporting the Fused Exact SwitchGLOBE manuscript.

Keywords: supplementary material; UAV routing; inference latency; reproducibility

1. Artifact and Provenance Inventory

Table 1 maps each claim family to its authoritative local artifact. Paths are relative to the project root. The paper copies publication-facing tables and figure sources into `paper/fused_exact_switchglobe`; the immutable experiment folders remain the source of record.

Table 1. Evidence inventory.

Claim family	Authoritative artifact	Scope and caveat
Eight-method routing comparison	R1	112,000 episode rows; five seeds; 14 scenarios; scenario-macro statistics
Exact-Fast reliability	R2	Paired seed differences; predeclared 0.5 percentage-point non-inferiority floor
A100 latency	R3	Same A100 session; batch 1; warm-up 50; 2,000 repeats; five checkpoints
Operator attribution	R4	Seed 42; held-out medium; 30 profiled calls; not the primary latency sample
Early-exit behavior	R5	43,467 decisions; 14,000 paired episodes; empirical, not formal, equivalence
Method implementation	R6	Canonical alias SwitchGlobe-Policy; fused decide() path
Literature corpus	R7	45 accessible Notion database records; metadata audit, not meta-analysis

Received:

Accepted:

Published:

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Artifact keys. R1: `artifacts/final_paper_simulation/synthesis/combined_comparison/`; R2: `artifacts/final_paper_simulation/full/ablation/validation/fast_vs_exact_overall.csv`; R3: `artifacts/switchglobe_latency_optimization/globev2_colab_cli_20260905/results/`; R4: `the sibling profile/results/ directory`; R5: `docs/latency_optimization/final_report.m`

d; R6: implementations/lite_globe/models/student_policy.py; R7: paper/fused_exact_s
witchglobe/literature_comparison.csv.

The combined manifest reports completion, 112,000 episode rows, 560 seed summaries, and exact cross-experiment row matching. It also records that the source worktree contained dirty files when the archive was synthesized. That disclosure is retained as a limitation. Before public release, the authors should rebuild the final archive from a clean tag and publish the commit, environment lock, and hashes together.

2. Metric Definitions and Statistical Unit

The primary reliability metric is connected-pair packet delivery ratio (PDR), which conditions on source–destination pairs that were initially connected. Overall PDR retains all episodes and therefore mixes routing failure with initial disconnection. Deadline delivery counts packets delivered within the simulator deadline. Successful-delay metrics condition on delivery; cells with no delivered packets are undefined and must be displayed as N/A rather than zero.

The simulator energy quantity is a proxy, not joules. The scenario-macro synthesis includes its documented failure treatment, whereas another pooled analysis uses a different weighting. Results in the manuscript and this supplement consistently follow the synthesis files. Policy-input bytes measure input tensor memory, not HELLO messages, route requests, eBPF update bytes, or total radio overhead.

The seed is the inferential unit. Individual episodes and individual timing repetitions reveal distributions but do not replace independent training seeds. With five seed pairs, a two-sided exact sign-flip test has minimum attainable $p = 0.0625$; effect size and confidence interval are therefore reported alongside the exact test.

3. Complete Figure Portfolio

Each figure below has a publication PNG and vector SVG in the supplementary-figure directory. Its plotted values are stored in a correspondingly numbered CSV in figure_data. Captions distinguish local CPU, current A100, older A100 evidence, full reliability, and diagnostic-only episode timing.

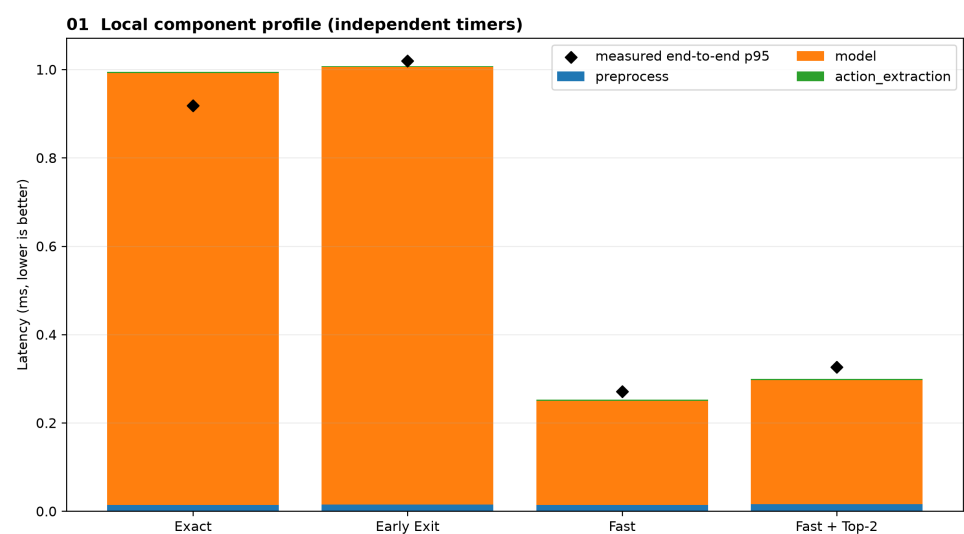


Figure 1. Independent component timers for the local batch-one policy path over five seeds. Components are measured separately for bottleneck diagnosis and must not be summed to recreate the end-to-end timer.

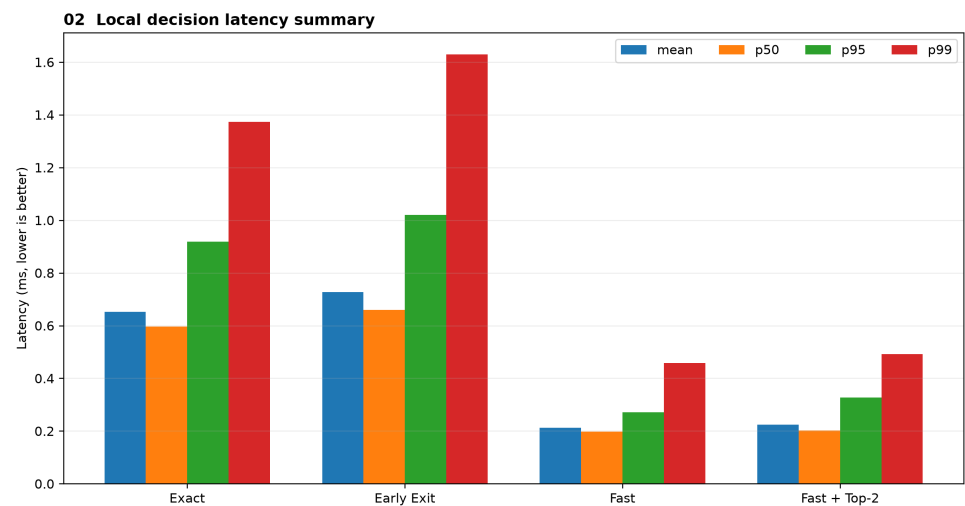


Figure 2. Mean, median, p95, and p99 latency summaries for exact and candidate variants. Fast variants are approximate policies and require a separate reliability judgment.

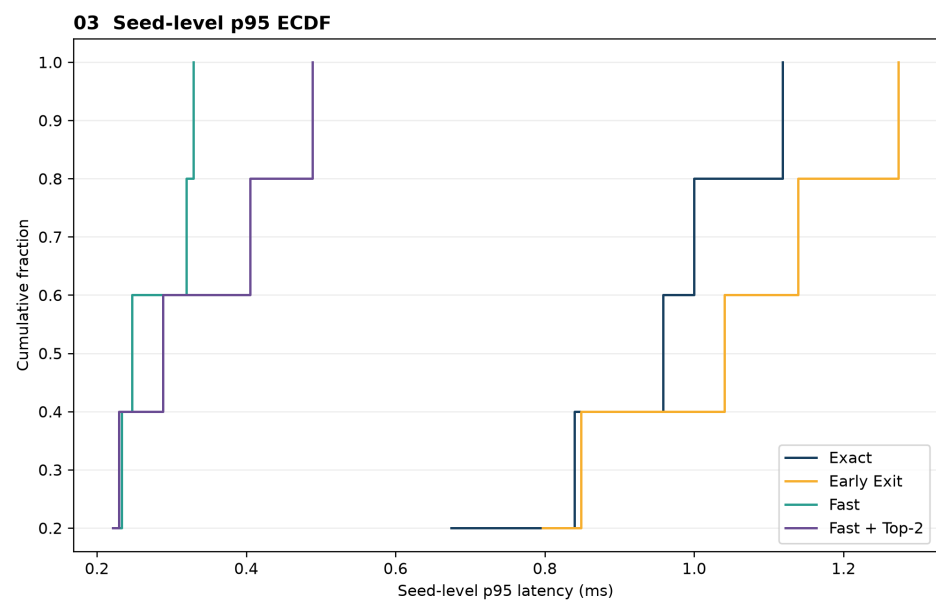


Figure 3. Empirical distribution of five seed-specific p95 values. It avoids presenting thousands of correlated routing decisions as independent experimental replications.

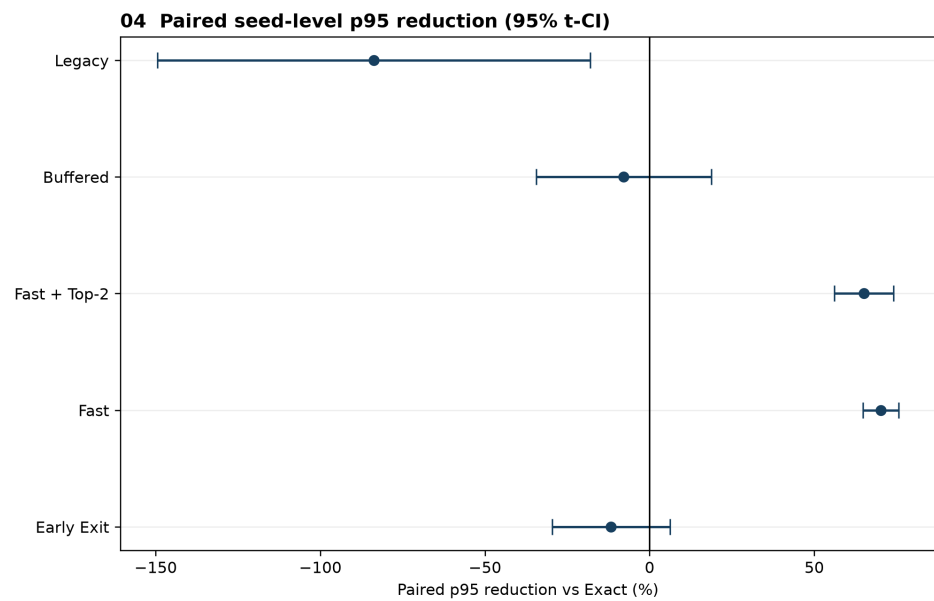


Figure 4. Paired p95 speed effects. Positive is faster than fused Exact. Confidence intervals are computed across matched checkpoints.

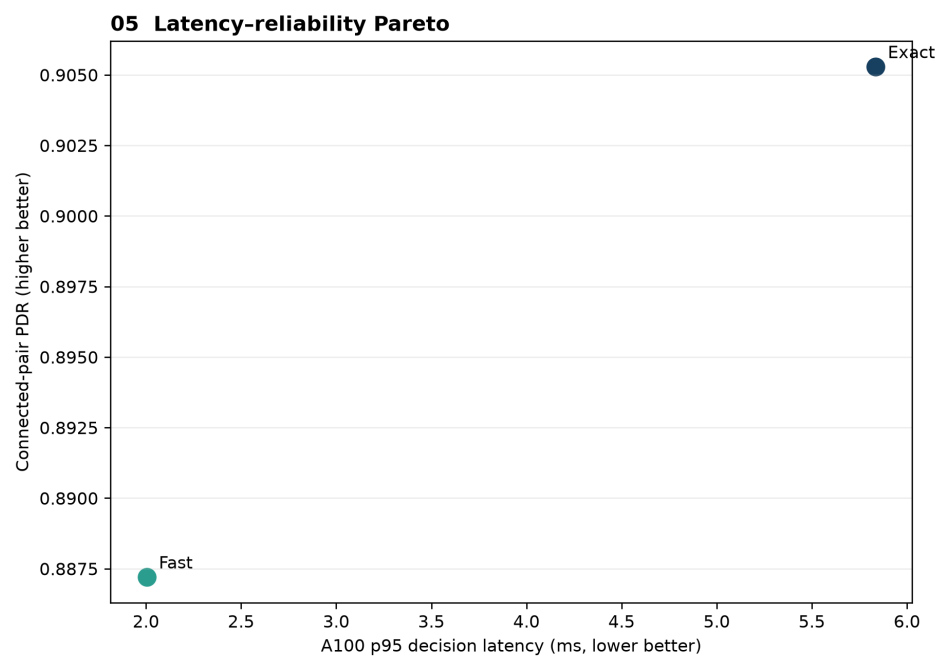


Figure 5. Connected-pair PDR versus A100 decision latency. Reliability and latency originate from separate, explicitly scoped experiments: the full routing evaluation and the same-session microbenchmark.

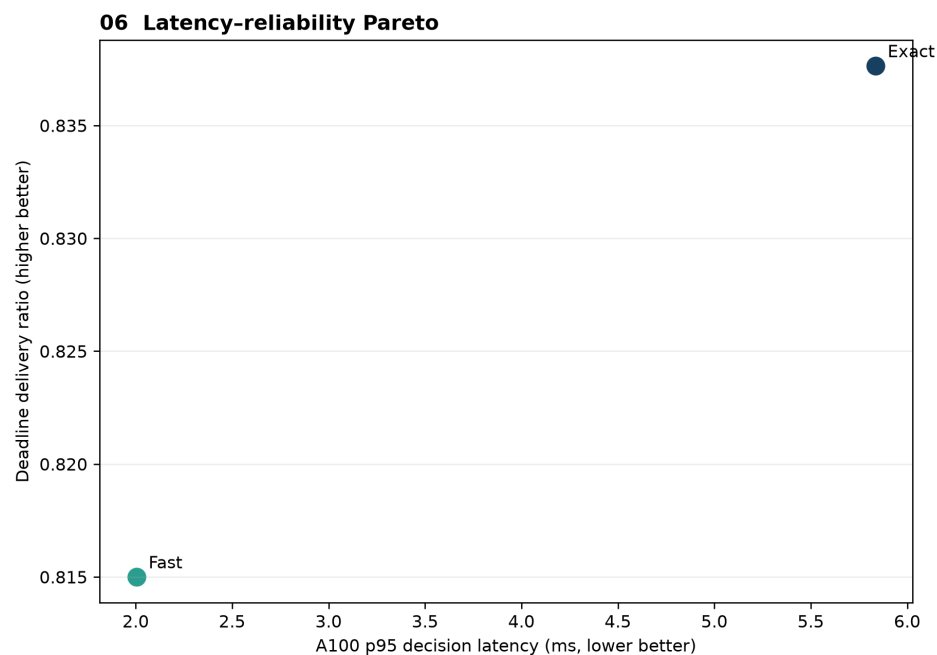


Figure 6. Deadline-delivery ratio versus A100 p95 decision latency. Exact occupies the reliability-oriented point; Fast occupies a lower-latency but lower-reliability point.

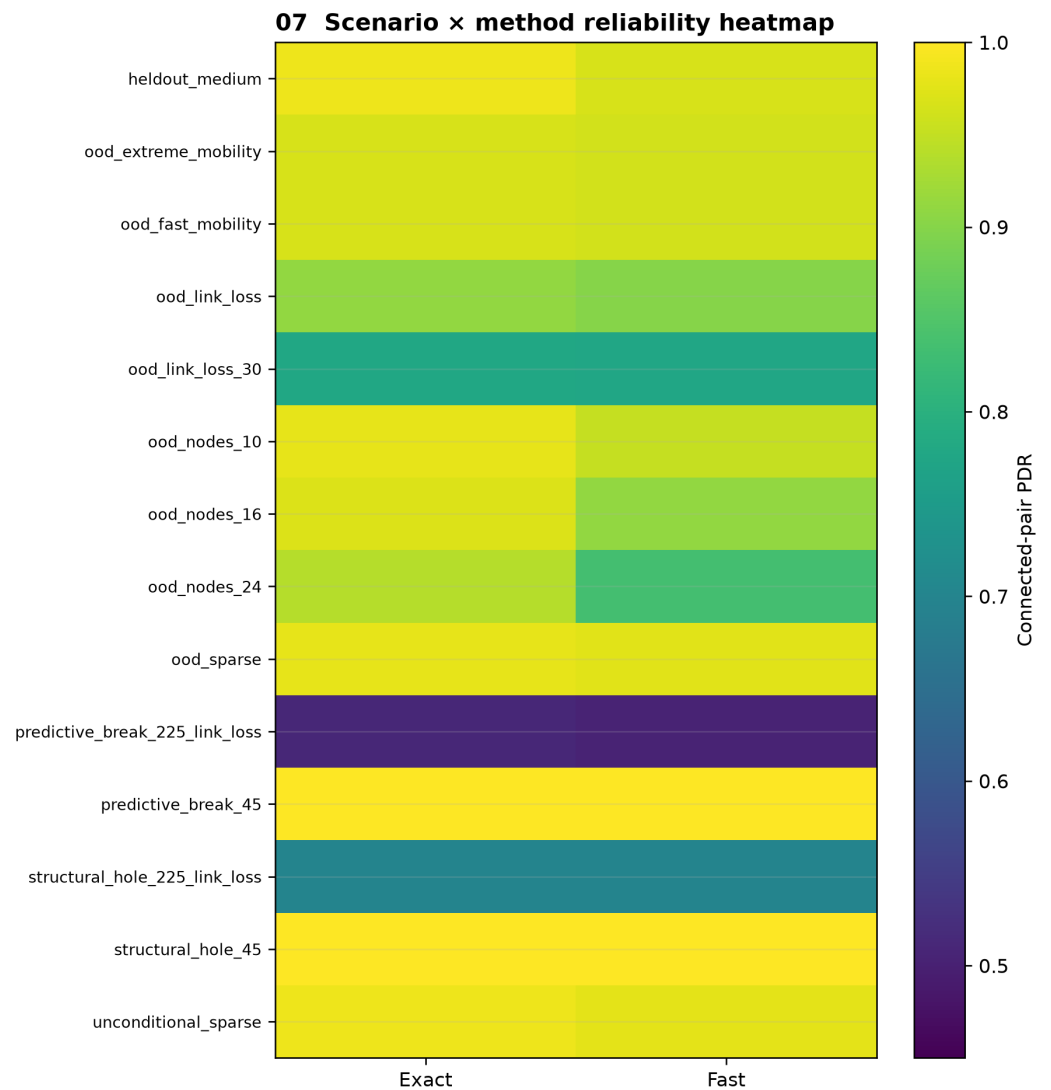


Figure 7. Scenario-level reliability heatmap. It reveals that Fast degradation is concentrated in specific OOD node-density and predictive-break conditions rather than uniform across the benchmark.

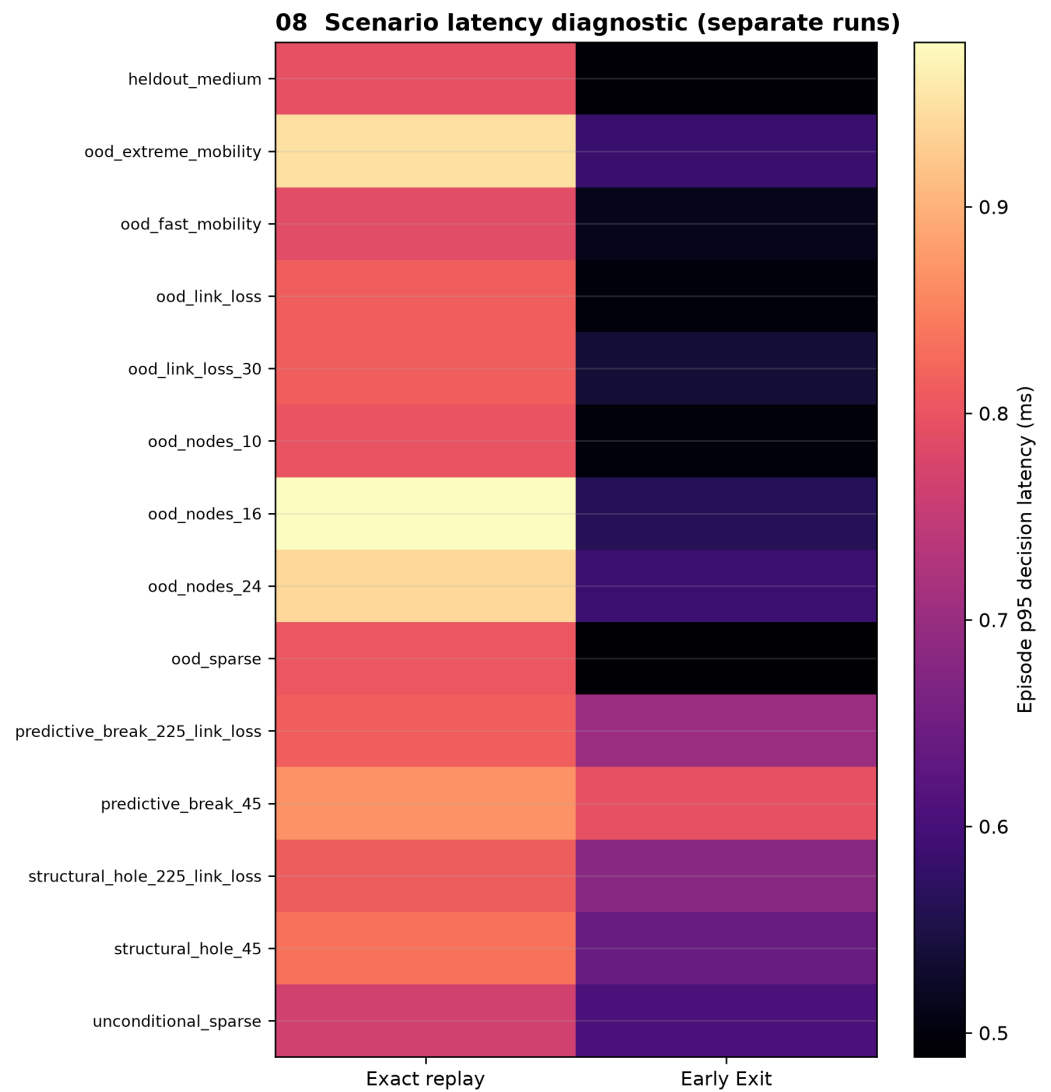


Figure 8. Diagnostic latency collected during the full 14,000-episode routing execution. It is useful for scenario localization but is not a same-session randomized microbenchmark and is not used as the primary A100 estimate.

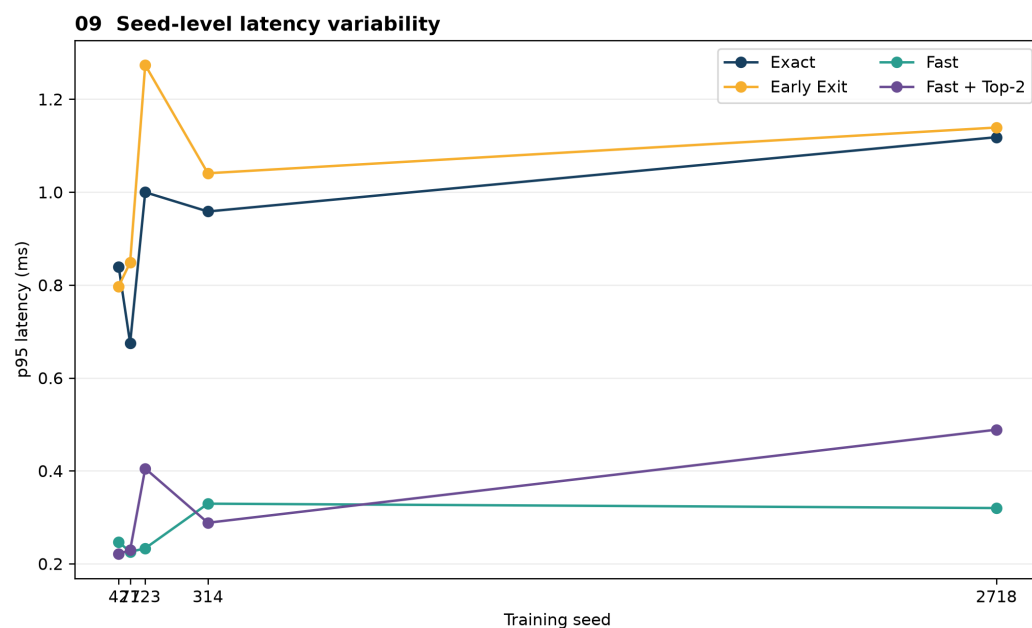


Figure 9. Seed-level variability for all optimized candidates. Matching seeds across variants enables paired rather than unpaired effect estimates.

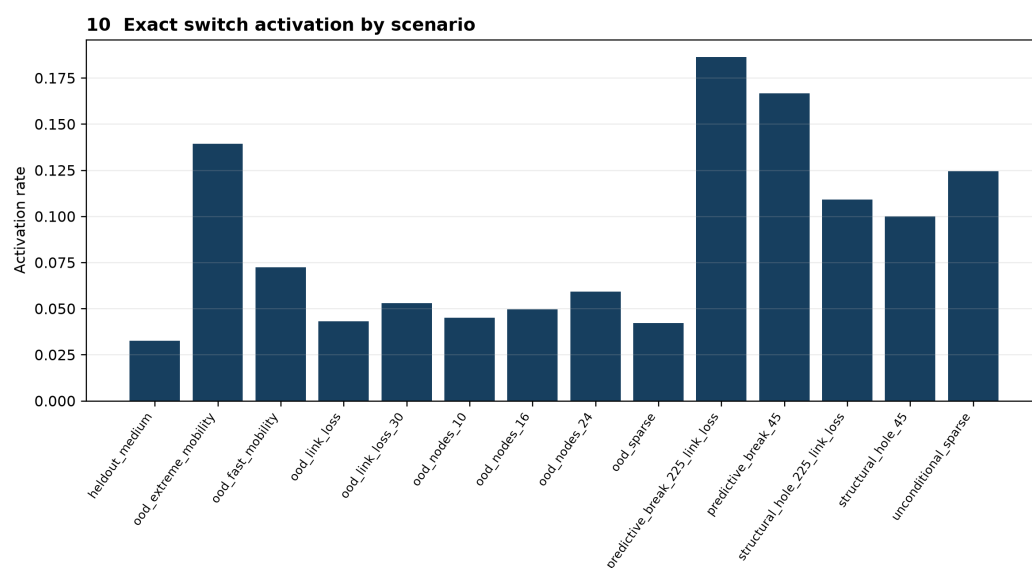


Figure 10. Switch activation by scenario. A switched action and the need to compute the predictive branch are different events: Exact computes both branches, while Early Exit attempts to skip a branch before the switch can be fully evaluated.

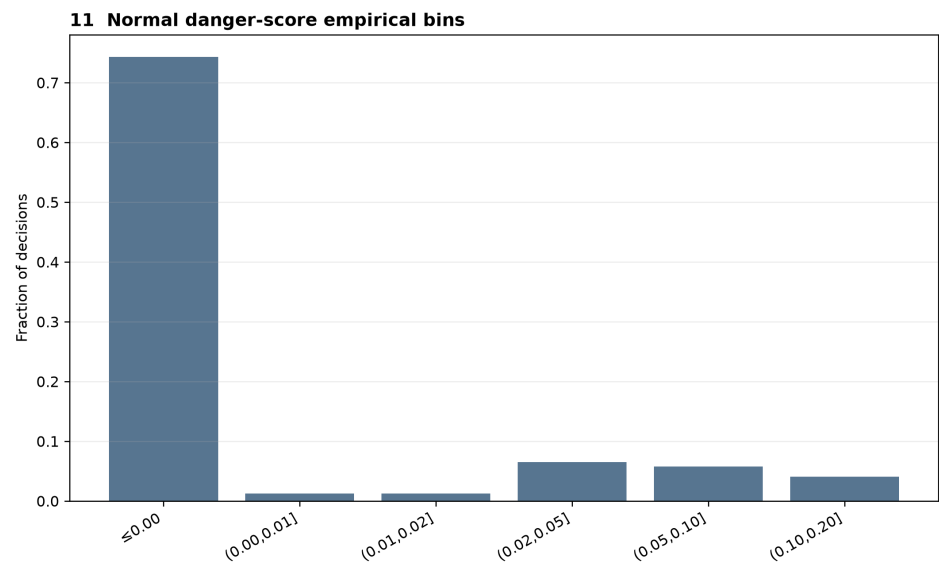


Figure 11. Gate-score interval mass reconstructed from cumulative sweep counts. It is not a raw continuous-score histogram, a distinction needed for reproducible interpretation.

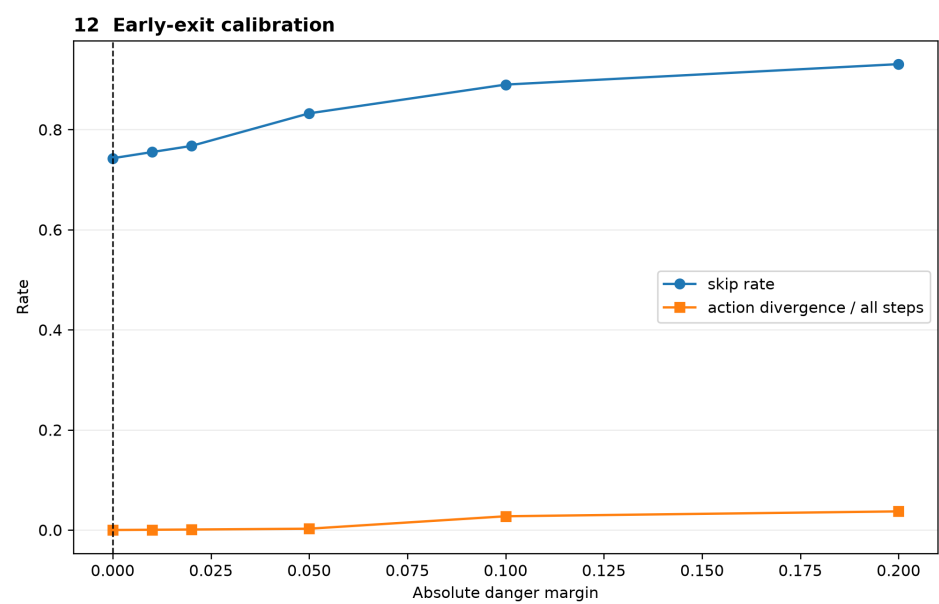


Figure 12. Early-exit margin sweep. Margin zero is the only tested value with zero observed action divergence over the complete decision trace; positive margins trade additional skipping for mismatches.

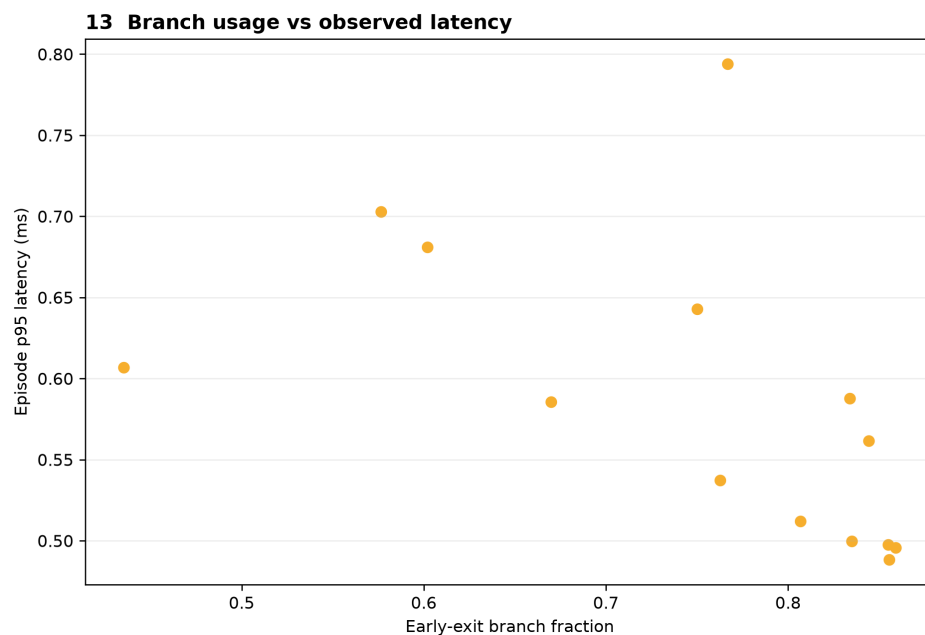


Figure 13. Branch-use rate versus measured latency. A 74.27% skip rate does not reduce p95 because Python control flow and synchronization exceed the small predictive-branch compute saved.

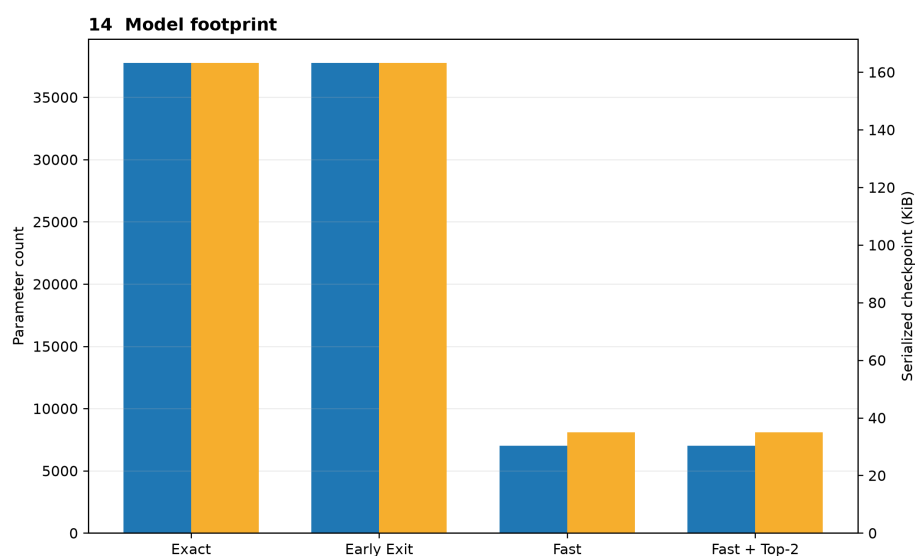


Figure 14. Parameter and serialized-size comparison. Early Exit shares the Exact weights; Fast contains 7,011 parameters and is a separately distilled approximate model.

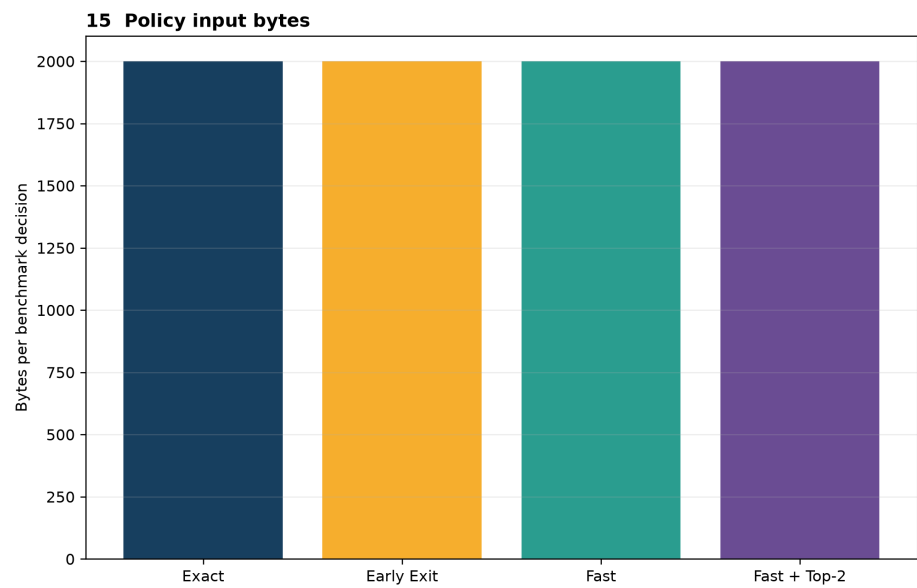


Figure 15. Policy-input memory footprint. This is not radio control overhead; feature acquisition traffic should be instrumented independently on a network implementation.

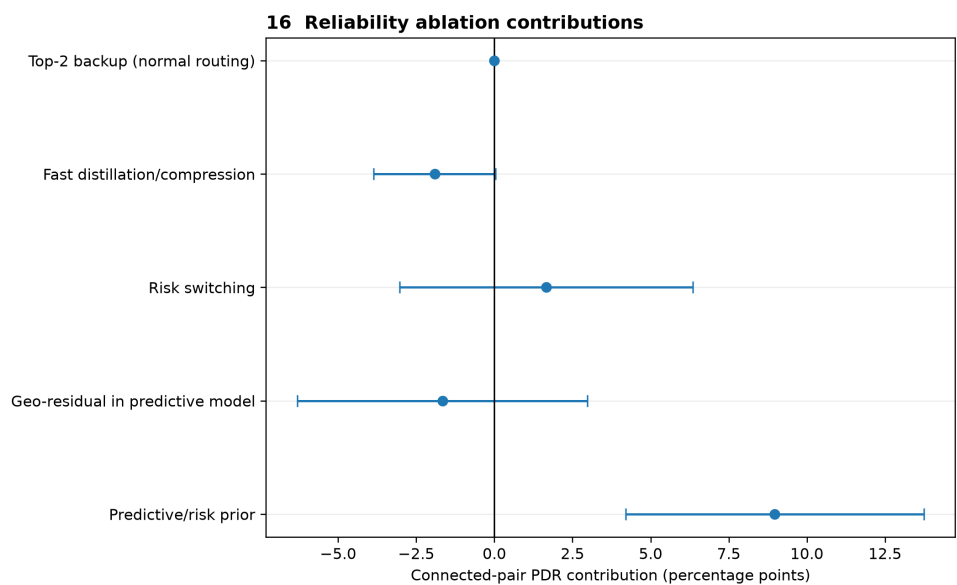


Figure 16. Directional ablation contributions with five-seed intervals. The plot separates the normal distilled policy, predictive information, and switching behavior.

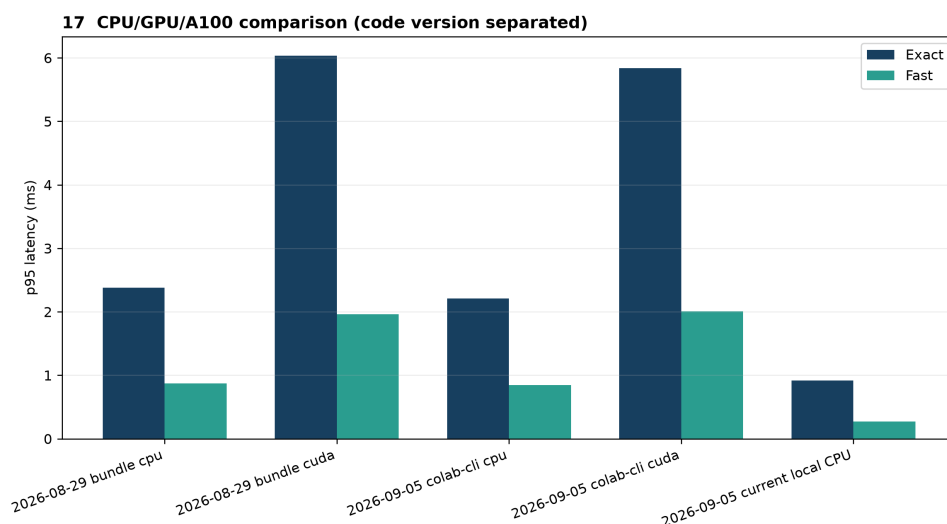


Figure 17. Device comparison. The current 2026-09-05 Colab-CLI run is primary; an older A100 bundle is shown only as an independent reproducibility reference and is not pooled with the current run.

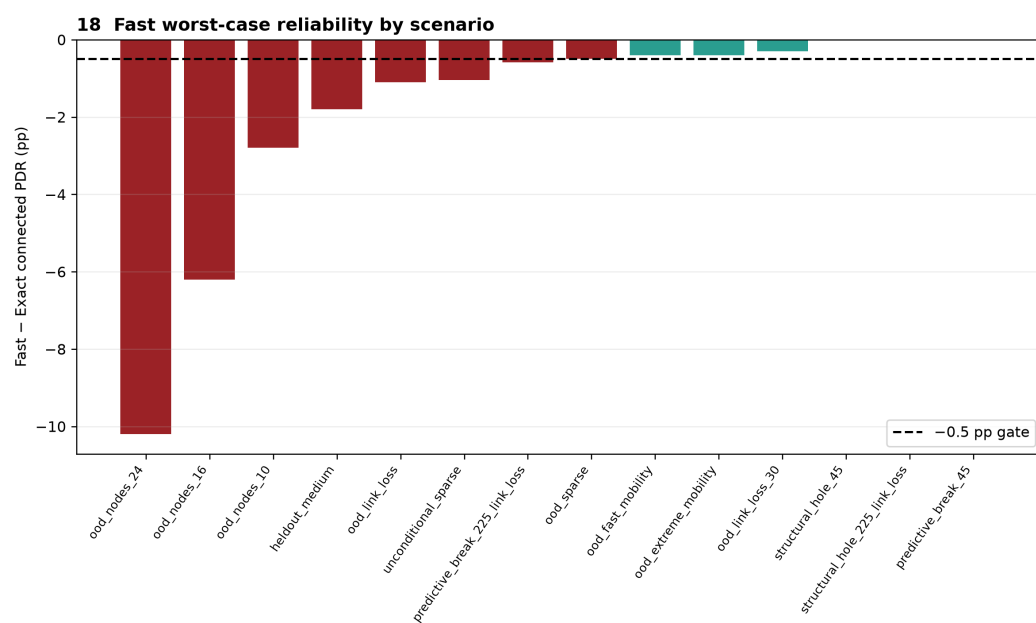


Figure 18. Worst-scenario localization. The figure prevents an acceptable macro average from hiding unacceptable OOD or link-loss behavior.

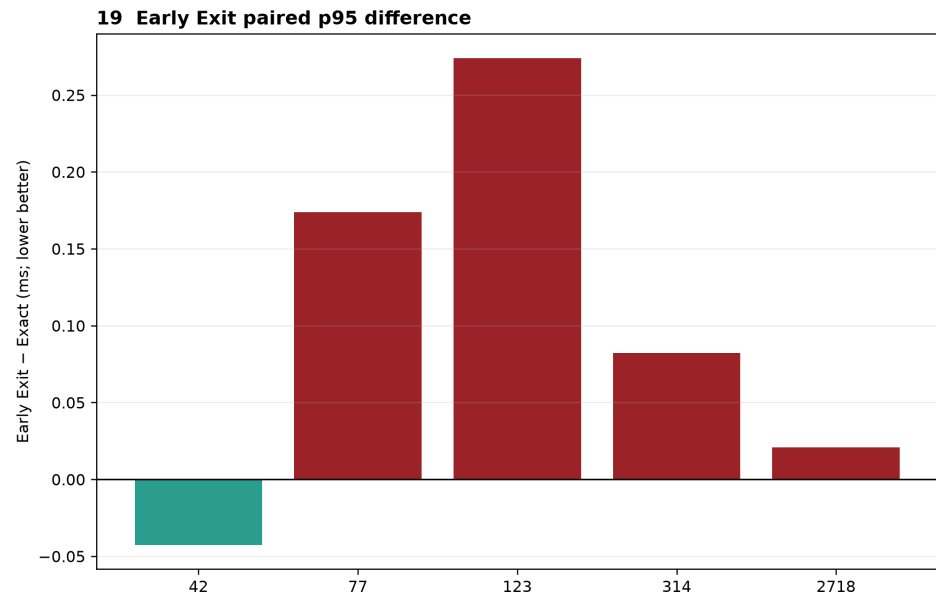


Figure 19. Early-Exit paired latency differences across five seeds. The average direction is worse and improvement is not consistent.

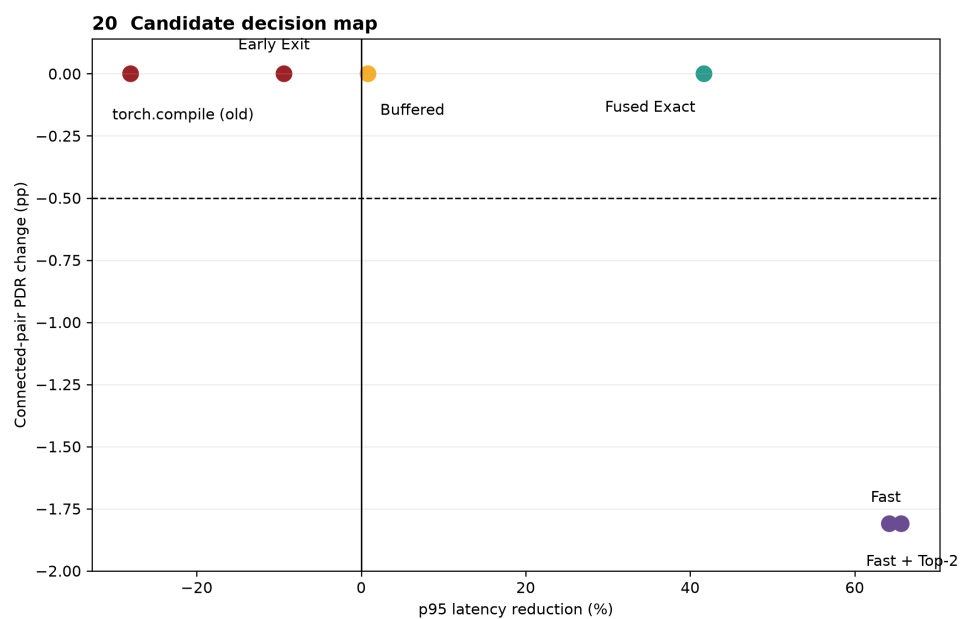


Figure 20. Candidate decision matrix. Fused Exact improves the duplicated legacy execution without an outcome change; buffering is too small; Early Exit is slower; Fast violates the quality gate.

4. Latency Tables

Table 2. Primary A100 session summary (five checkpoint means).

Variant	Device	Mean	p50	p95	p99	Decision
Fused Exact	CPU	2.168	2.161	2.208	2.344	retain
Early Exit	CPU	2.366	2.356	2.427	2.581	reject
Fast	CPU	0.812	0.814	0.847	0.927	quality fail
Fast + Top-2	CPU	0.853	0.852	0.881	0.933	quality un-resolved
Buffered Exact	CPU	2.187	2.179	2.244	2.370	hold
Legacy repeated	CPU	3.698	3.687	3.775	3.935	superseded
Fused Exact	CUDA	5.671	5.648	5.836	6.101	retain
Early Exit	CUDA	6.198	6.172	6.386	6.663	reject
Fast	CUDA	1.957	1.950	2.005	2.120	quality fail
Fast + Top-2	CUDA	2.040	2.032	2.092	2.183	quality un-resolved
Buffered Exact	CUDA	5.626	5.602	5.788	6.020	hold
Legacy repeated	CUDA	9.741	9.706	10.004	10.376	superseded

Table 3. Paired CUDA p95 effects relative to Fused Exact. Positive reduction means faster.

Candidate	p95 reduction	95% <i>t</i> CI	Bootstrap 95% CI	Interpretation
Early Exit	−9.42%	[−10.62, −8.22]	[−10.22, −8.61]	slower
Buffered Exact	+0.83%	[+0.21, +1.44]	[+0.48, +1.25]	too small; CPU reverses
Fast	+65.65%	[+65.06, +66.24]	[+65.24, +65.95]	latency pass, quality fail
Fast + Top-2	+64.16%	[+63.65, +64.66]	[+63.84, +64.50]	latency pass, quality unresolved

5. Exact–Fast Reliability Gate

Table 4. Fast minus Exact directional paired effects. Higher-is-better metrics use Fast–Exact; lower-is-better metrics use Exact–Fast, so negative always means degradation.

Metric	Exact	Fast	Directional change	95% CI	Gate
Connected-pair PDR	0.90529	0.88720	−0.01809	−0.03606 to −0.00011	fail
Deadline ratio	0.83764	0.81500	−0.02264	−0.04625 to +0.00096	fail
p95 success delay	4.2643	4.4864	−0.2221	−0.4208 to −0.0235	fail
Energy/delivered proxy	2.2279	2.3694	−0.1415	−0.2799 to −0.0031	fail

The connected-PDR interval lies entirely below zero and its lower bound is far below the allowed −0.005. Fast therefore cannot be called statistically equivalent or non-inferior to Exact. The deadline interval overlaps zero at its upper edge, but its lower edge independently violates the gate. The delay and energy directions reinforce the rejection.

6. Operator-Profile Interpretation

The Exact profile contains 1,050 `aten::_local_scalar_dense` events and the same number of device-to-host copies over 30 calls. Early Exit contains 1,140 of each; Fast contains 300; Fast+Top-2 contains 330. The counts are evidence of synchronization frequency,

not time percentages for the primary benchmark. Exact also shows Boolean reductions such as `aten::all` and `aten::any`, which are consistent with action-mask and gate logic.

This evidence motivates three implementation hypotheses: (i) preserve gate and action extraction as tensors until the final API boundary; (ii) replace Python branching with a runtime-native conditional graph; and (iii) fuse small elementwise and reduction operations where the deployment compiler can do so without altering masks or threshold comparisons. Any optimized graph must pass exact logit/action replay or the same routing-quality gate used for Fast.

7. Reproduction Checklist

1. Check out the recorded source commit and verify that no untracked implementation changes remain.
2. Verify SHA-256 for the source bundle and five Exact/Fast checkpoint archives.
3. Run the full routing evaluation with seeds 42, 77, 123, 314, and 2718; 14 scenario definitions; and 200 episodes per cell.
4. Verify 14,000 episodes per method, 112,000 combined rows, finite metrics, valid actions, and mask compliance.
5. Aggregate per seed and scenario before forming 14-scenario macro values; preserve N/A successful-delay cells.
6. Run CPU and CUDA latency variants in one session, one CPU thread, batch one, 50 warm-ups, and 2,000 repetitions per checkpoint.
7. Synchronize CUDA before and after the end-to-end timing region; keep cold start separate.
8. Compute matched-seed effects, Student-*t* intervals, deterministic bootstrap intervals, exact sign-flip tests, and Holm-adjusted comparisons.
9. Run operator profiling separately and do not merge profiler timings with primary latency values.
10. Regenerate all figures from the copied CSV sources and compare the figure manifest.

8. Reporting Checklist for Submission

Before submission, the authors must replace anonymous metadata, verify the current Drones template version, insert a public repository/DOI or a defensible restricted-access statement, add the precise simulator license, state the compute resource provider, and decide whether adapted baseline code can be released. A target-device experiment is strongly recommended. If it cannot be completed, the title, abstract, and conclusion must retain the term “deployment-oriented” rather than “deployed” or “real-time onboard.”